STAT 6206: Bayesian Statistics with Applications in Machine Learning
Assignment 1 - Solution

Academic year 25/26, second term

1. (a) Up to proportionality, the likelihood in θ has the form

$$p(y \mid \theta) \;\propto\; \theta^{2a}\exp(-\theta^2 y^2).$$

Let $\phi = \theta^2$ ($\phi > 0$). Then

$$p(y \mid \phi) \;\propto\; \phi^{a}\exp(-\phi y^2),$$

which is the kernel part of Gamma distribution. Hence a conjugate prior for ϕ is Gamma:

$$\phi = \theta^2 \sim Gamma(\alpha, \beta),\ \alpha > 0,\ \beta > 0.$$

Transforming back to θ using $\phi = \theta^2$ and $d\phi = 2\theta\, d\theta$,

$$p(\theta) = p(\phi)\cdot\left|\frac{d\phi}{d\theta}\right| = \frac{\beta^\alpha}{\Gamma(\alpha)}(\theta^2)^{\alpha-1}e^{-\beta\theta^2}\cdot 2\theta = \frac{2\beta^\alpha}{\Gamma(\alpha)}\,\theta^{2\alpha-1}e^{-\beta\theta^2}, \qquad \theta > 0.$$

1. (b) Let $Y_1, \ldots, Y_n \overset{iid}{\sim}$ Galenshore(a, θ) and define

$$S = \sum_{i=1}^{n} y_i^2.$$

The joint likelihood (in θ) is

$$p(y_{1:n} \mid \theta) \;\propto\; \theta^{2an}\exp\left(-\theta^2\sum_{i=1}^{n} y_i^2\right) = \theta^{2an}e^{-\theta^2 S}.$$

In terms of $\phi = \theta^2$, this becomes

$$p(y_{1:n} \mid \phi) \;\propto\; \phi^{an}e^{-\phi S}.$$

With prior $\phi \sim Gamma(\alpha, \beta)$, the posterior is

$$p(\phi \mid y_{1:n}) \propto \phi^{\alpha-1}e^{-\beta\phi}\cdot\phi^{an}e^{-S\phi} = \phi^{\alpha+an-1}e^{-(\beta+S)\phi},$$

hence

$$\phi = \theta^2 \mid y_{1:n} \sim Gamma(\alpha + an,\ \beta + S)\ .$$

Equivalently, the posterior density of θ is

$$p(\theta \mid y_{1:n}) = \frac{2(\beta+S)^{\alpha+an}}{\Gamma(\alpha+an)}\,\theta^{2(\alpha+an)-1}\exp\big(-(\beta+S)\theta^2\big), \qquad \theta > 0.$$

1. (c) Let $k = \alpha + an$ and $r = \beta + S$. Since $\phi \mid y_{1:n} \sim Gamma(k, r)$ and $\theta = \sqrt{\phi}$, we use the Gamma moment

$$\mathbb{E}[\phi^{1/2}] = r^{-1/2}\frac{\Gamma(k+1/2)}{\Gamma(k)}.$$

Therefore,

$$\mathbb{E}[\theta \mid y_{1:n}] = (\beta+S)^{-1/2}\frac{\Gamma(\alpha+an+\frac{1}{2})}{\Gamma(\alpha+an)}\ .$$

Let $Y_1, \ldots, Y_n \overset{iid}{\sim} \text{Poisson}(\lambda)$ with $\lambda > 0$. Denote $y_{1:n} = (y_1, \ldots, y_n)$ and $S = \sum_{i=1}^n y_i$.

2. (a) The likelihood function is

$$L(\lambda \mid y_{1:n}) = \prod_{i=1}^n \frac{e^{-\lambda}\lambda^{y_i}}{y_i!} = e^{-n\lambda}\lambda^S \prod_{i=1}^n \frac{1}{y_i!}.$$

Hence the log-likelihood is

$$\ell(\lambda) = \log L(\lambda \mid y_{1:n}) = -n\lambda + S\log\lambda - \sum_{i=1}^n \log(y_i!).$$

Differentiate and set to zero:

$$\frac{d}{d\lambda}\ell(\lambda) = -n + \frac{S}{\lambda} = 0 \quad \Longrightarrow \quad \hat{\lambda}_{\text{MLE}} = \frac{S}{n} = \bar{y}.$$

2. (b) Assume a Gamma prior in *shape–rate* form:

$$\lambda \sim \Gamma(a, b),\ a > 0,\ b > 0,$$

i.e.

$$p(\lambda) = \frac{b^a}{\Gamma(a)}\lambda^{a-1}e^{-b\lambda}, \qquad \lambda > 0.$$

The posterior density is proportional to likelihood times prior:

$$p(\lambda \mid y_{1:n}) \propto L(\lambda \mid y_{1:n})\, p(\lambda) \propto \left(e^{-n\lambda}\lambda^S\right)\left(\lambda^{a-1}e^{-b\lambda}\right) = \lambda^{a+S-1}e^{-(b+n)\lambda}.$$

Therefore,

$$\lambda \mid y_{1:n} \sim Gamma(a+S,\ b+n)$$

The posterior mean and variance are

$$\mathbb{E}[\lambda \mid y_{1:n}] = \frac{a+S}{b+n}\ , \qquad \text{Var}(\lambda \mid y_{1:n}) = \frac{a+S}{(b+n)^2}\ .$$

3. (a) The posterior distribution $p(\mu \mid y)$ is

$$\begin{aligned} p(\mu \mid y) &\propto \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^N (y_i-\mu)^2\right)\cdot\exp\left(-\frac{1}{2\tau^2}(\mu-\mu_0)^2\right) \\ &= \exp\left\{-\frac{1}{2}\left(\frac{N}{\sigma^2}+\frac{1}{\tau^2}\right)\mu^2 + \left(\frac{\sum_{i=1}^N y_i}{\sigma^2}+\frac{\mu_0}{\tau^2}\right)\mu\right\}. \end{aligned}$$

Thus,

$$\mu \mid y \sim \mathcal{N}(\mu_N, \sigma_N^2),$$

where

$$\sigma_N^2 = \left(\frac{N}{\sigma^2} + \frac{1}{\tau^2}\right)^{-1}, \qquad \mu_N = \sigma_N^2 \left(\frac{\sum_{i=1}^N y_i}{\sigma^2} + \frac{\mu_0}{\tau^2}\right).$$

Equivalently, letting $\bar{y} = \frac{1}{N}\sum_{i=1}^N y_i$, we may write

$$\mu_N = \sigma_N^2 \left(\frac{N\bar{y}}{\sigma^2} + \frac{\mu_0}{\tau^2}\right).$$

3. (b) The posterior predictive distribution for a new observation $\tilde{Y}$ is given by:

$$\begin{aligned}
p(\tilde{Y} \mid y) &= \int p(\tilde{Y} \mid \mu)\, p(\mu \mid y)\, d\mu \\
&= \int \mathcal{N}(\tilde{Y} \mid \mu, \sigma^2)\, \mathcal{N}(\mu \mid \mu_N, \sigma_N^2)\, d\mu \\
&= \int \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\tilde{Y}-\mu)^2}{2\sigma^2}\right) \cdot \frac{1}{\sqrt{2\pi\sigma_N^2}} \exp\left(-\frac{(\mu-\mu_N)^2}{2\sigma_N^2}\right) d\mu \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \frac{1}{\sqrt{2\pi\sigma_N^2}} \int \exp\left(-\frac{(\tilde{Y}-\mu)^2}{2\sigma^2} - \frac{(\mu-\mu_N)^2}{2\sigma_N^2}\right) d\mu \\
&\propto \int \exp\left(-\frac{1}{2}\Big(\frac{1}{\sigma^2} + \frac{1}{\sigma_N^2}\Big)\mu^2 + \Big(\frac{\tilde{Y}}{\sigma^2} + \frac{\mu_N}{\sigma_N^2}\Big)\mu\right) d\mu \\
&\propto \exp\left(\frac{\big(\frac{\tilde{Y}}{\sigma^2} + \frac{\mu_N}{\sigma_N^2}\big)^2}{2\big(\frac{1}{\sigma^2} + \frac{1}{\sigma_N^2}\big)}\right) \exp\left(-\frac{\tilde{Y}^2}{2\sigma^2} - \frac{\mu_N^2}{2\sigma_N^2}\right) \\
&\propto \exp\left(-\frac{1}{2(\sigma^2 + \sigma_N^2)}(\tilde{Y} - \mu_N)^2\right).
\end{aligned}$$

Thus, the posterior predictive distribution is

$$\tilde{Y} \mid y \sim \mathcal{N}\big(\mu_N,\ \sigma^2 + \sigma_N^2\big).$$

3. (c) Since

$$\tilde{Y} \mid y \sim \mathcal{N}\big(\mu_N,\ \sigma^2 + \sigma_N^2\big),$$

the posterior predictive mean is

$$\mathbb{E}[\tilde{Y} \mid y] = \mu_N.$$

A 95% posterior predictive credible interval for $\tilde{Y}$ is

$$\mathrm{CI}_{0.95} = \left(\mu_N - z_{0.975}\sqrt{\sigma^2 + \sigma_N^2},\ \mu_N + z_{0.975}\sqrt{\sigma^2 + \sigma_N^2}\right),$$

where $z_{0.975}$ denotes the 97.5th percentile of the standard normal distribution.

4. (a) **Priors.** Let $\beta = (\beta_0, \ldots, \beta_p)^\top \in \mathbb{R}^{p+1}$ and $u = (u_1, \ldots, u_n)^\top$. A standard choice is:

$$\beta \sim \mathcal{N}(b_0, B_0), \qquad u_i \mid \sigma_u^2 \overset{iid}{\sim} \mathcal{N}(0, \sigma_u^2), \qquad \sigma_u^2 \sim \mathrm{IG}(a_0, b_0),$$

where $b_0 \in \mathbb{R}^{p+1}$, $B_0 \in \mathbb{R}^{(p+1)\times(p+1)}$ is positive definite, and $a_0, b_0 > 0$ are fixed hyperparameters.

4. (b) **Full conditional distributions.** Define the linear predictor

$$\eta_{it} = x_{it}^\top \beta + u_i, \qquad p_{it} = \mathrm{logit}^{-1}(\eta_{it}) = \frac{1}{1 + e^{-\eta_{it}}}.$$

Ignoring factors not depending on the parameters (e.g. $\binom{m_{it}}{y_{it}}$), the Binomial likelihood can be written as

$$p(y \mid \beta, u) \;\propto\; \prod_{i=1}^{n}\prod_{t=1}^{T} p_{it}^{y_{it}}(1-p_{it})^{m_{it}-y_{it}} = \exp\left(\sum_{i=1}^{n}\sum_{t=1}^{T}\left[y_{it}\eta_{it} - m_{it}\log(1+e^{\eta_{it}})\right]\right).$$

(i) Full conditional for β. Combining the likelihood with the normal prior on β yields

$$p(\beta \mid u, \sigma_u^2, y, x) \propto \exp\left(\sum_{i=1}^{n}\sum_{t=1}^{T}\left[y_{it}\eta_{it} - m_{it}\log(1+e^{\eta_{it}})\right] - \frac{1}{2}(\beta - b_0)^\top B_0^{-1}(\beta - b_0)\right),$$

where $\eta_{it} = x_{it}^\top \beta + u_i$. This full conditional is *not* of a standard conjugate form.

(ii) Full conditional for each u_i. For a fixed i,

$$p(u_i \mid \beta, \sigma_u^2, y, x) \propto \exp\left(\sum_{t=1}^{T}\left[y_{it}\eta_{it} - m_{it}\log(1+e^{\eta_{it}})\right] - \frac{u_i^2}{2\sigma_u^2}\right),$$

where $\eta_{it} = x_{it}^\top \beta + u_i$. This full conditional is also non-conjugate.

(iii) Full conditional for σ_u^2. Using $u_i \mid \sigma_u^2 \sim \mathcal{N}(0, \sigma_u^2)$ and $\sigma_u^2 \sim \mathrm{IG}(a_0, b_0)$,

$$p(\sigma_u^2 \mid u) \propto (\sigma_u^2)^{-(a_0+1)} \exp\left(-\frac{b_0}{\sigma_u^2}\right) \cdot (\sigma_u^2)^{-n/2} \exp\left(-\frac{\sum_{i=1}^n u_i^2}{2\sigma_u^2}\right),$$

hence

$$\sigma_u^2 \mid u \sim \mathrm{IG}\left(a_0 + \frac{n}{2},\; b_0 + \frac{1}{2}\sum_{i=1}^{n} u_i^2\right).$$